

Introduction to Spatial Data

Outline

- Conceptual Models of Space
- Spatial Data Models
 - Vector
 - Topology
 - Raster
- Attribute Data
- Database Management

Conceptual Models of Space

- Objects present a view of the world in which the Earth surface is populated by a number discrete entities (e.g., houses, streets, lakes)
 - Objects can be 0-, 1-, 2-, or 3-dimensional
 - Object types include points, lines, polygons, networks

Conceptual Models of Space

- Fields present a view of the world in which Earth is a continuous surface that is characterized by a number of properties
 - e.g., temperature, land use, soil pH

Conceptual Models of Space

- Important differences between object and field conceptual models
 - Observed element
 - In the object model, focus is on the “object”
 - Object may have multiple attributes
 - In the field model, focus is on the “attribute”
 - Attribute varies over space
 - Empty space
 - In the object model, space between entities is “empty”
 - No empty space in the field model

Spatial Data Models

- We use spatial data models to describe the conceptual models for digital representation
 - Two primary models are vector and raster
- Vector (object)
 - Uses spatial “features”
 - Points, lines, polygons
- Raster (field)
 - Uses grid cells
 - Each cell coded with a single value

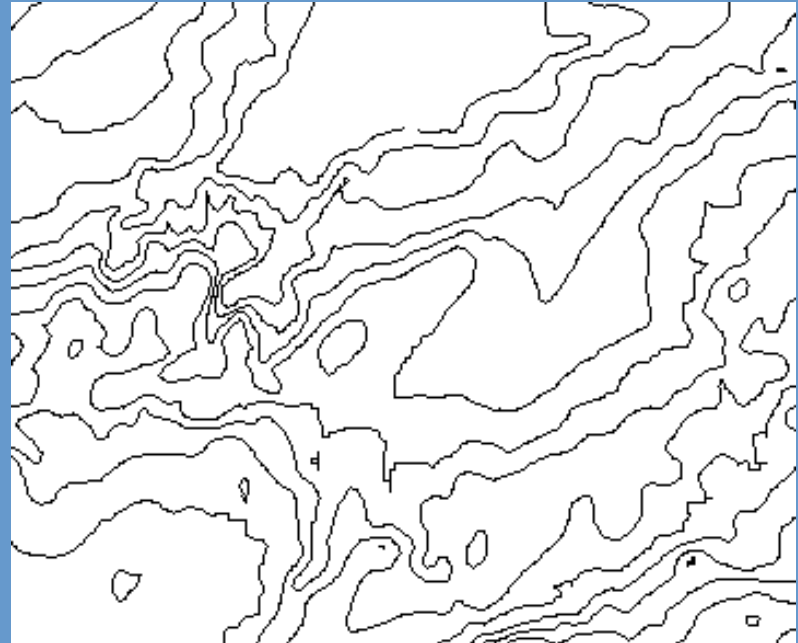
Spatial Data Models

- Consider how elevation can be represented

Grid of values

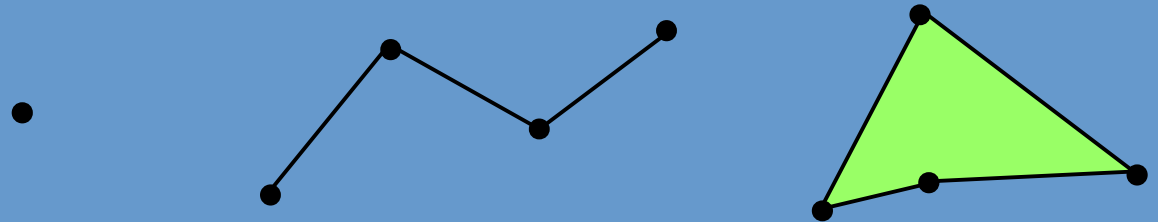


Contour lines



Vector Data

- Feature classes:
 - Points, Lines, Polygons
 - Points are the basic building blocks of vector data



- Generally, a vector data layer contains a single “feature class” or “type”
 - Data layers contain multiple features (or discrete objects)
 - Each object may have multiple attributes

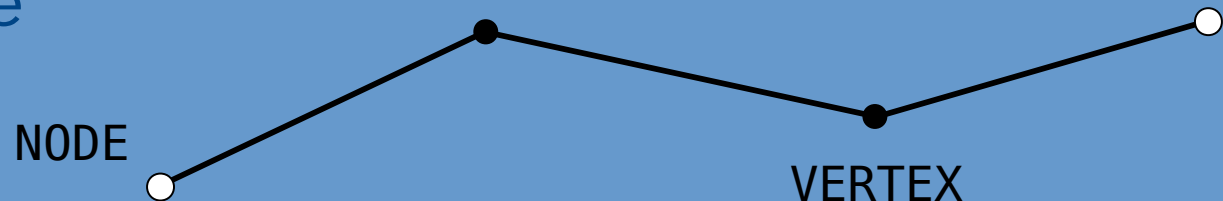
Vector Features

- Point

- A single coordinate pair of values (x,y)
- 0 dimensional object
 - No length or area
 - Important to recognize the difference between data representation and data display
 - Points have an “area” when we view them in the GIS
 - Control over the point size via the symbology
 - Data remains 0-D
 - Tuck this information away, because it will be useful to remember when we start talking about distance and GIS overlay

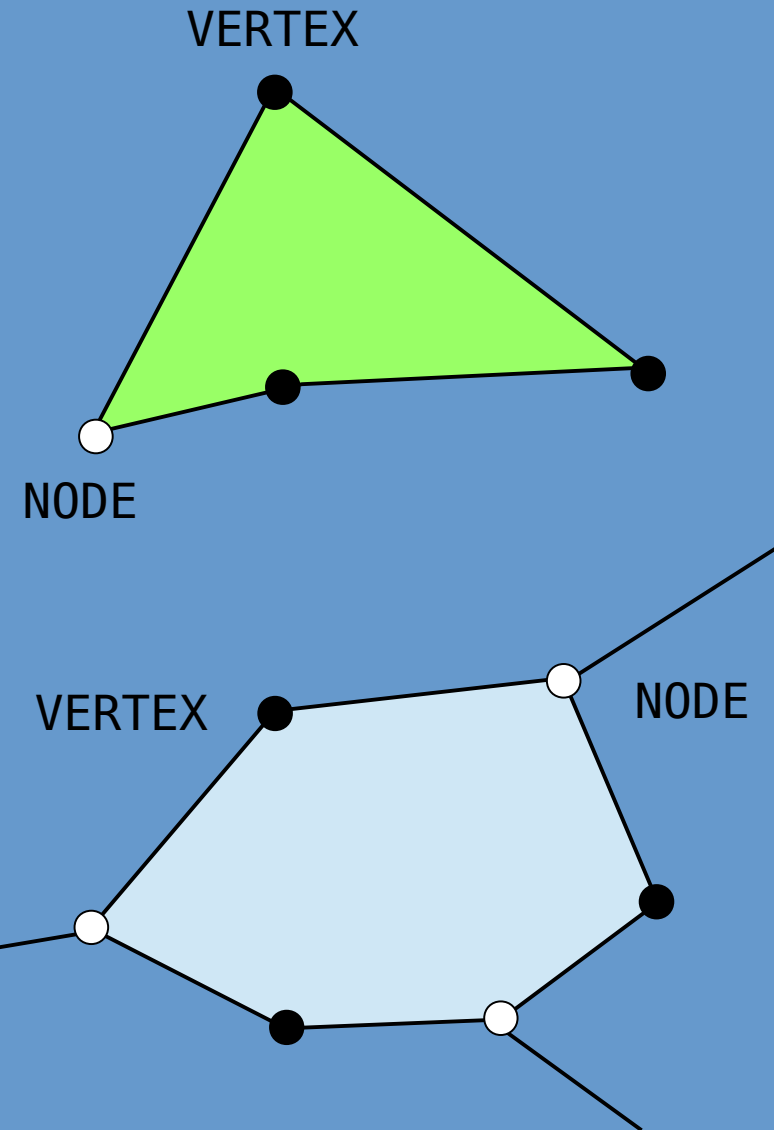
Vector Features

- Line
 - An ordered set of points
 - 1 dimensional object
 - Nodes
 - Starting and ending points
 - From node = Start point, To node = End point
 - Vertices
 - Intermediate points that define the shape of the line



Vector Features

- Polygon
 - A little more complex
 - 2 dimensional object
 - A single, closed line
 - Node
 - Starting and ending point(s)
 - Vertices
 - Intermediate points that define the shape of the line
 - Can be a set of connected lines forming an enclosed area



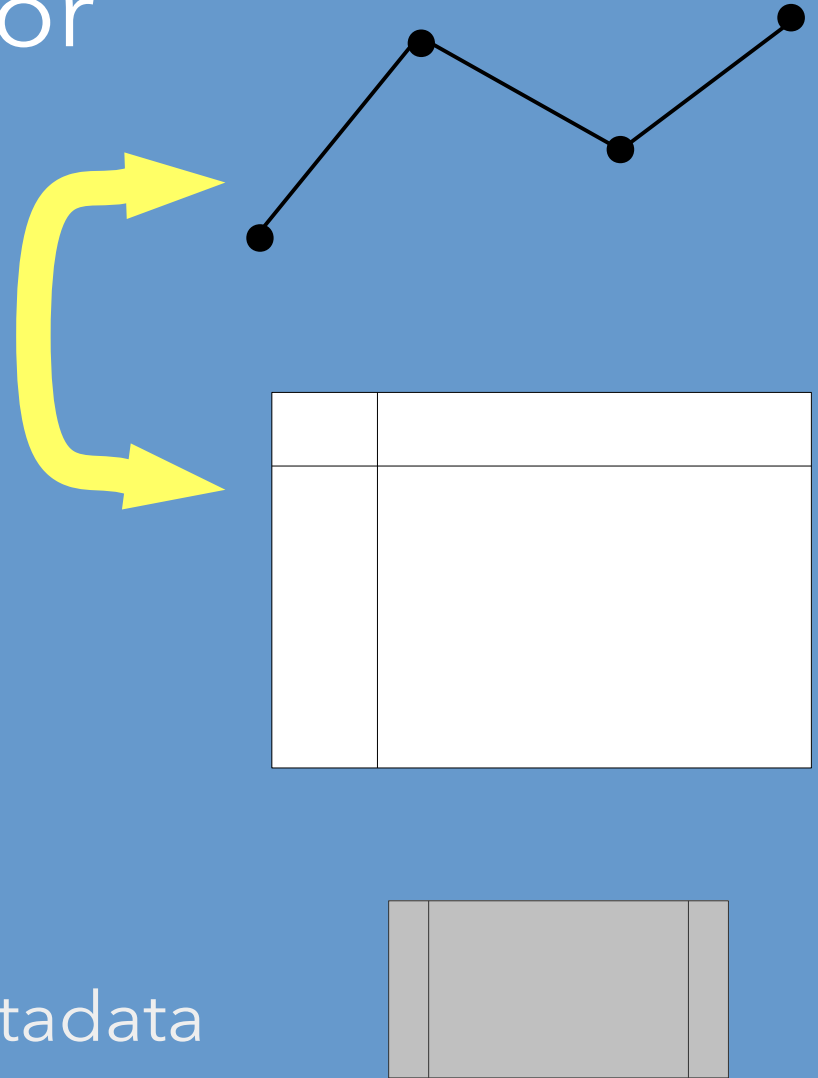
Vector Features

- Can you think of health related examples?
 - Points
 - Lines
 - Polygons

Basic Vector File Structure

- Multiple files per layer or feature class
 - Geometry
 - Spatial features
 - Tabular format
 - Attributes
 - Attribute information
 - Tabular format
 - Other
 - Indexes, projection info, metadata
 - File format varies (e.g., tables, text files)

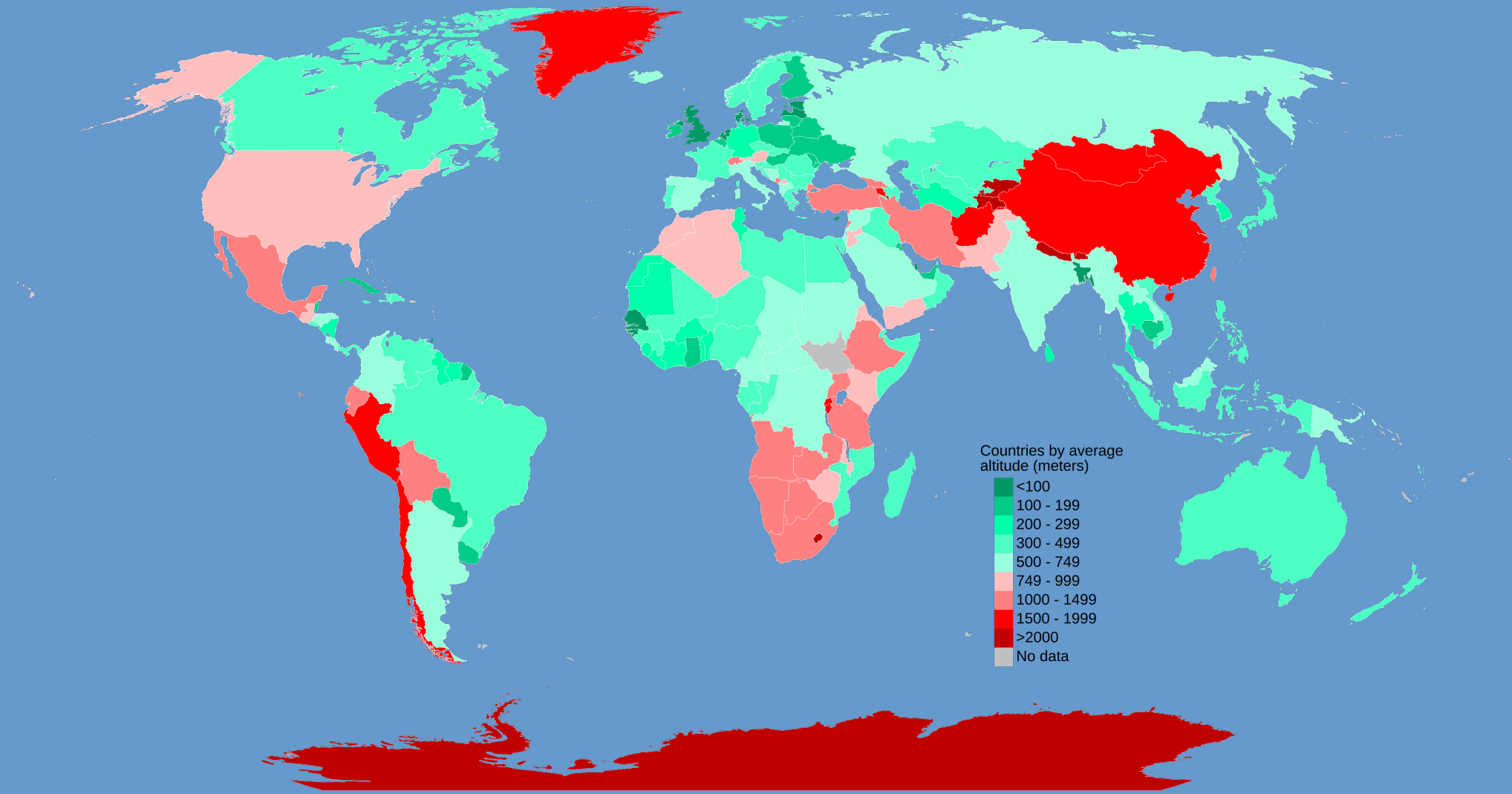
UNIQUE
ID



Field View

- For some phenomena or attributes of interest, field-type representation seems more appropriate
 - When phenomena or attribute varies continuously across space
 - Cases when discrete representation does not adequately capture the nature of the phenomena
 - E.g., Weather or climate, land use / land cover, air pollution, elevation

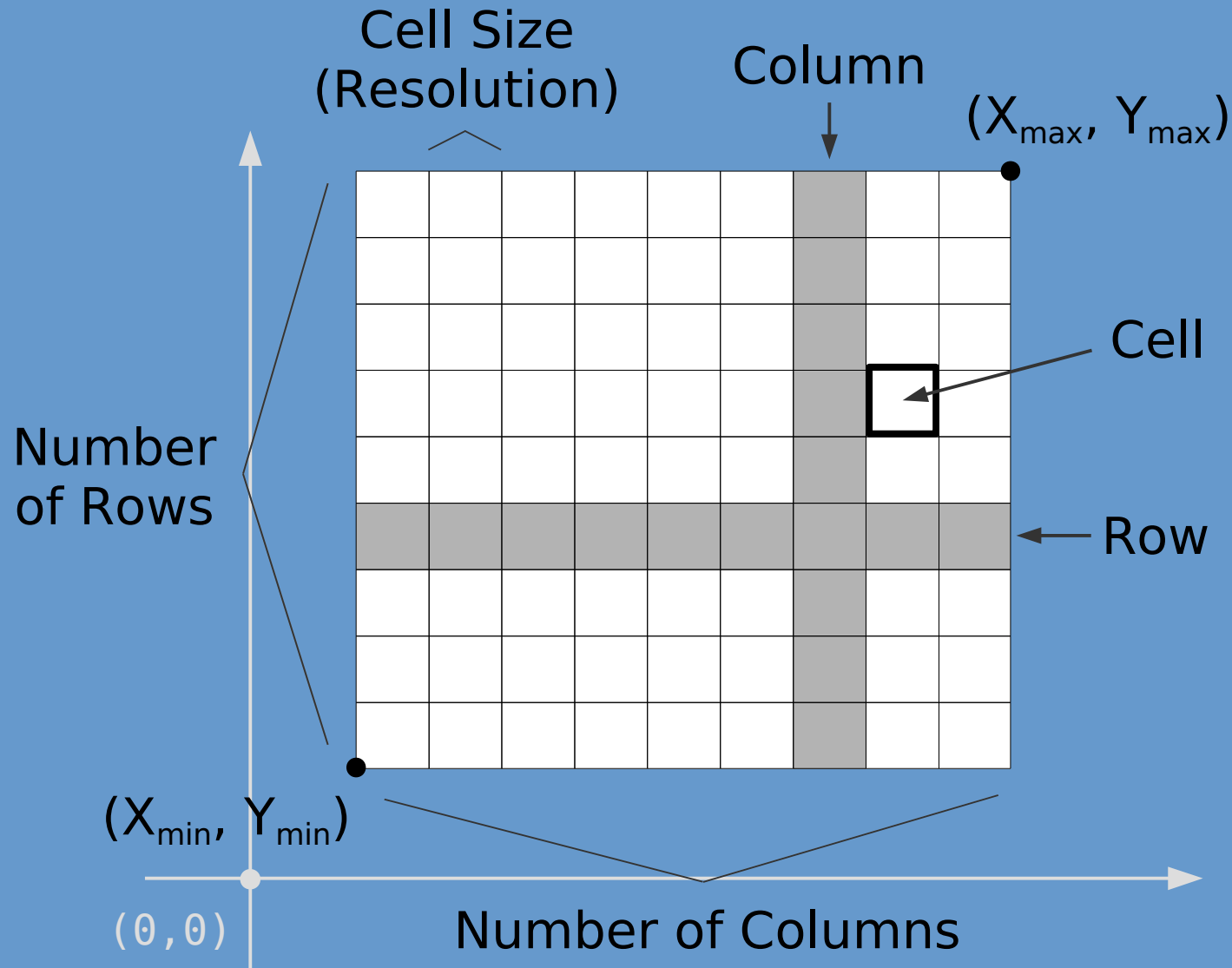
Field View



Field View

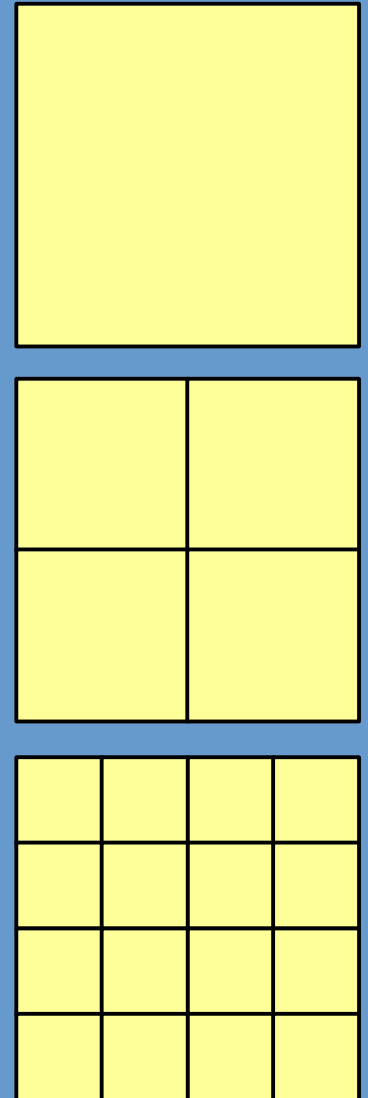
- Represented as raster data
 - Attribute based
 - Regularly spaced tessellations
 - Generally, square grid cells (or pixels)
 - Every cell location has a specific value

Raster Data Basics



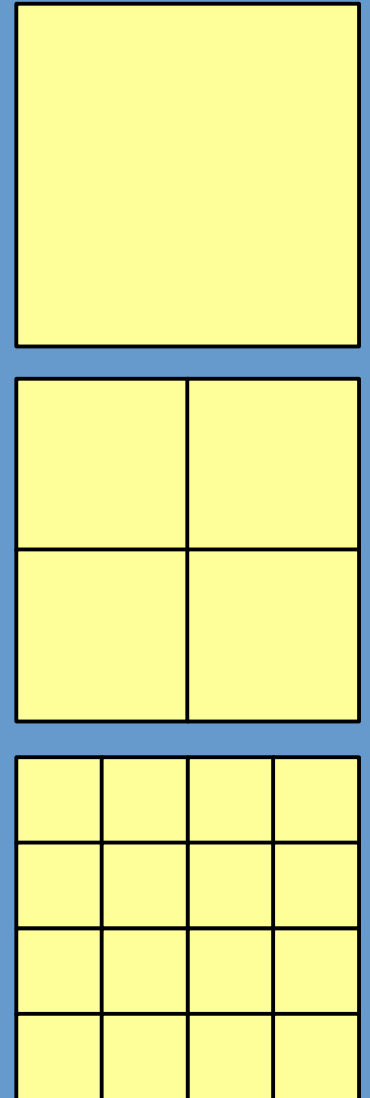
Cell Size

- Precision
 - $\frac{1}{2}$ of the cell size = 4x more detail in the grid
 - $\frac{1}{4}$ of the cell size = 16x more detail in the grid
- Cell size vs. file size
 - $\frac{1}{2}$ of the cell size = 4x more cells
 - $\frac{1}{4}$ of cell size = 16x more cells
 - Files can become quite large

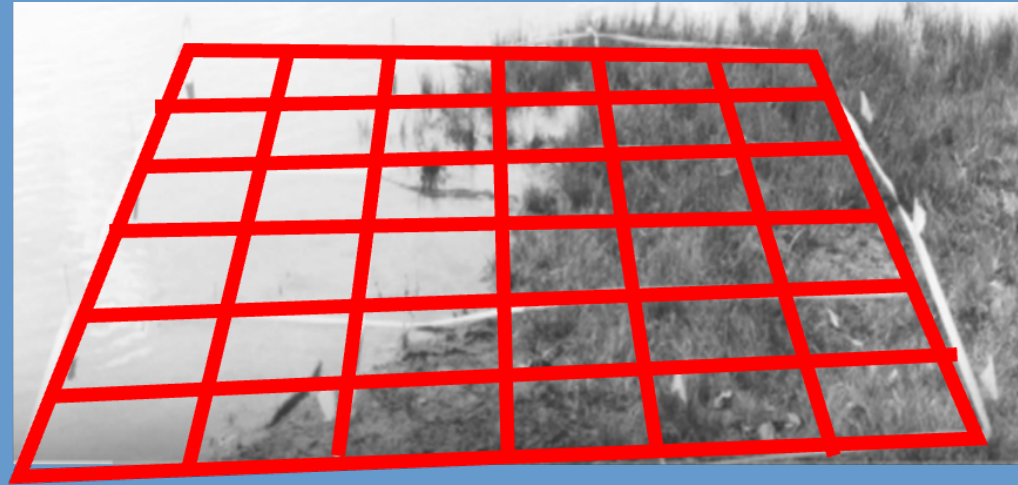
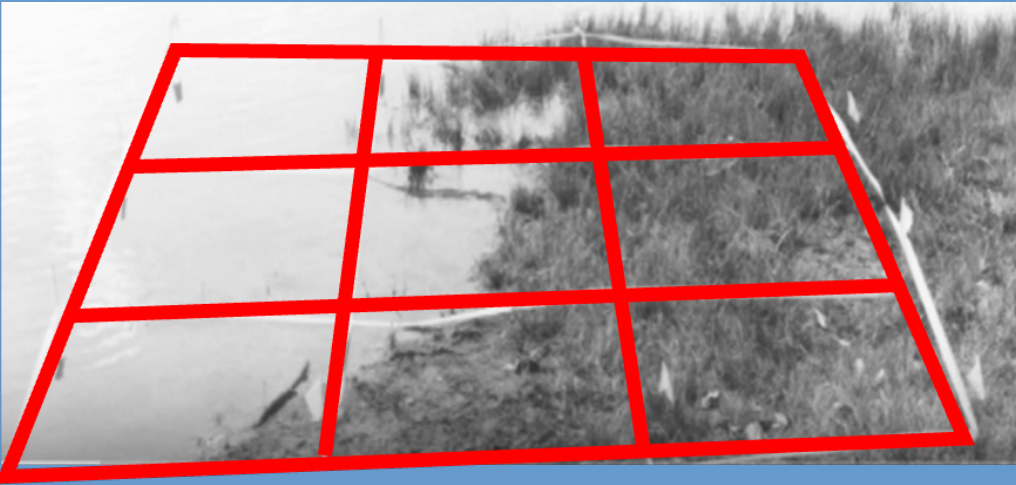


Cell Size

- Precision
 - $\frac{1}{2}$ of the cell size = 4x more detail in the grid
 - $\frac{1}{4}$ of the cell size = 16x more detail in the grid
- Cell size vs. file size
 - $\frac{1}{2}$ of the cell size = 4x more cells
 - $\frac{1}{4}$ of cell size = 16x more cells
 - Files can become quite large



Cell Size



Raster Data

- Analysis of multiple raster layers generally requires that the layers have similar:
 - Coordinate systems
 - Cell size
 - Cell locations
 - i.e., cells must align in space
- When using data from multiple sources, this can cause problems
 - One layer will have to be processed (spatially)
 - Resampling

Raster Attributes

- Data values (of the cells) are generally integers or real numbers
 - If the numbers correspond to specific categories or classes, these are generally stored in a separate table
 - e.g., extremely simplified landcover
 - 1 = bare soil, 2 = light vegetation, 3 = heavy vegetation

Other Data Formats

- Triangulated Irregular Networks (TINs)
 - Field-based structure for representing terrain
- Networks
 - Object-based structure for flow networks (roads, pipelines, rivers, etc.)
 - Very useful for travel analysis

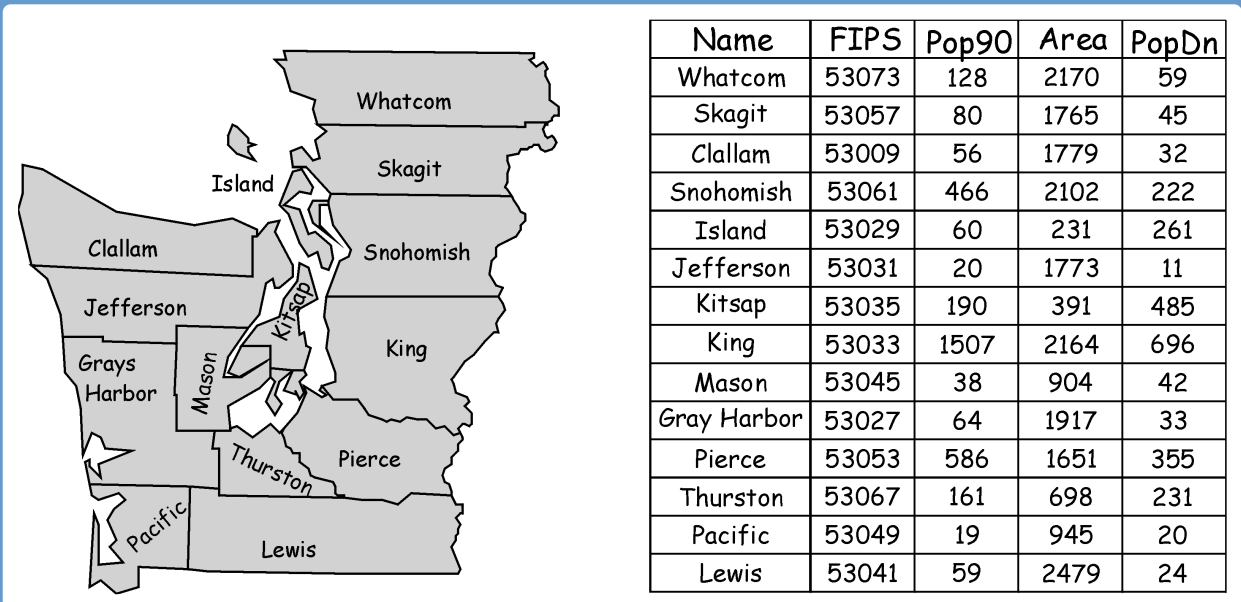
Why Important?

- Spatial processing/analysis capabilities and algorithms
 - Different for vector and raster data
 - Because of their different geometric representation
 - How your data are represented will influence the types of analysis that can be performed (and/or how you perform them)

Spatial Data

- Spatial data are stored in complex data formats (multiple files)
 - Geometry (spatial) and attributes (non-spatial)
 - Generally, geometry and attributes are stored in separate files

Generally, a concern for vector data



Attribute Files/Tables

- *Records* contain a set of attribute values for a single entity or feature
 - Table rows
- *Fields* are attributes or variables describing the features
 - Table columns

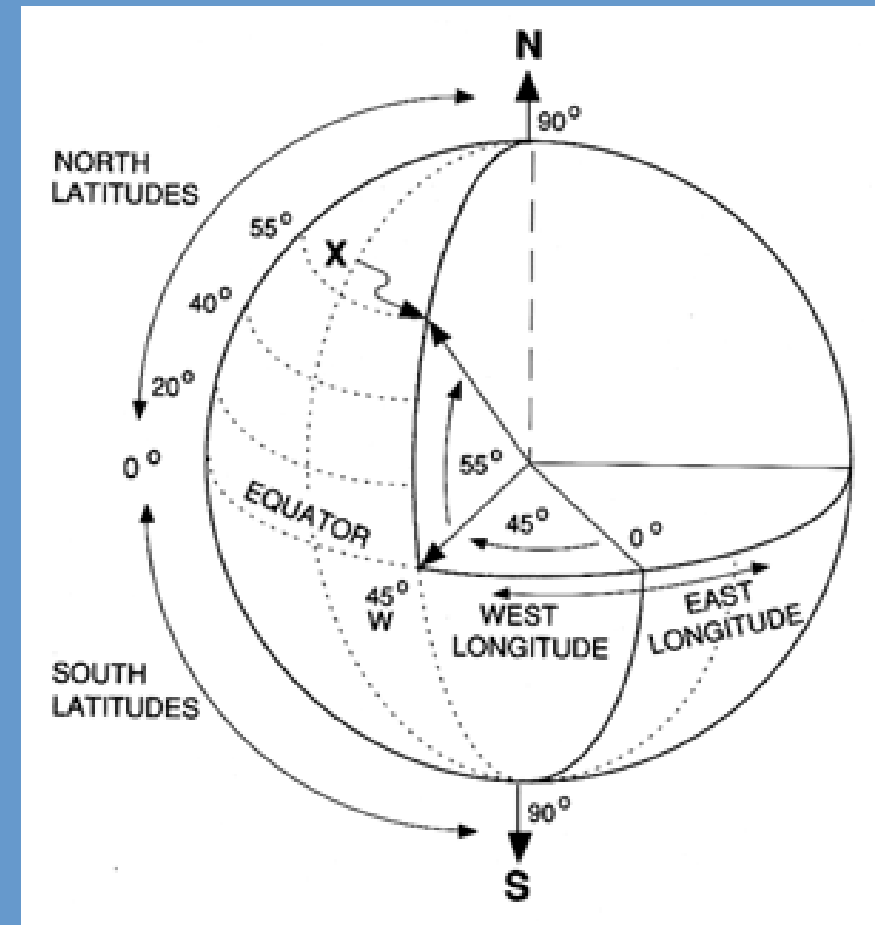
Record

S

Record #	PLACENAME	STATE	Field Pop 2010
1	Adrian	MI	22281
2	Albion	MI	10095
3	Alger	MI	2015
4	Algonac	MI	4612
5	Allegan	MI	4592
6	Allen Park	MI	31051
7	Allendale	MI	6950
8	Alma	MI	9075
9	Almont	MI	2375
10	Alpena	MI	11358
11	Ann Arbor	MI	109646
12	Atlanta	MI	2299
13	Atlantic Mine	MI	3379
14	Avoca	MI	2325

Geographic Coordinates

- On a sphere, referencing is made using angles, using the center of the Earth as the origin
- Latitude
 - Measured N-S relative to Equator
- Longitude
 - Measured E-W relative to Prime Meridian



Shape of the Earth

- Sphere

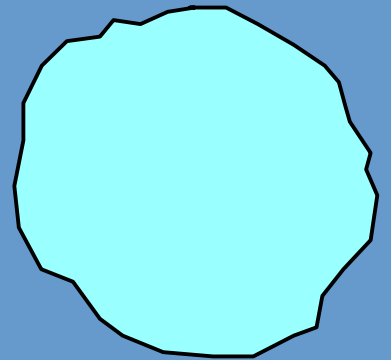
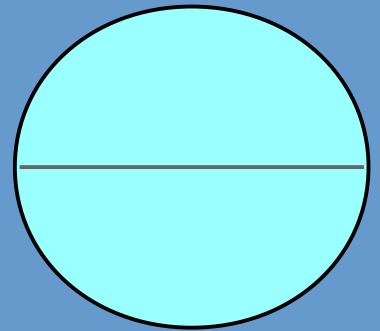
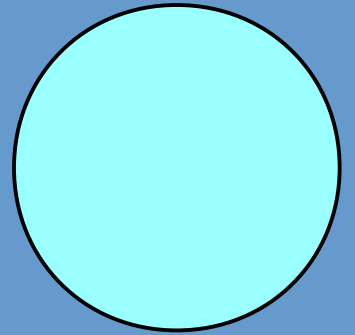
- Assumes perfectly spherical Earth
- Not realistic

- Ellipsoid

- Assumes flattening at poles, bulging at equator
- Described by “flattening ratio”

- Geoid

- Accounts for gravitational variations across Earth’s surface (i.e., variations in “sea level”)
 - An observed quantity that measures deviation from ellipsoid

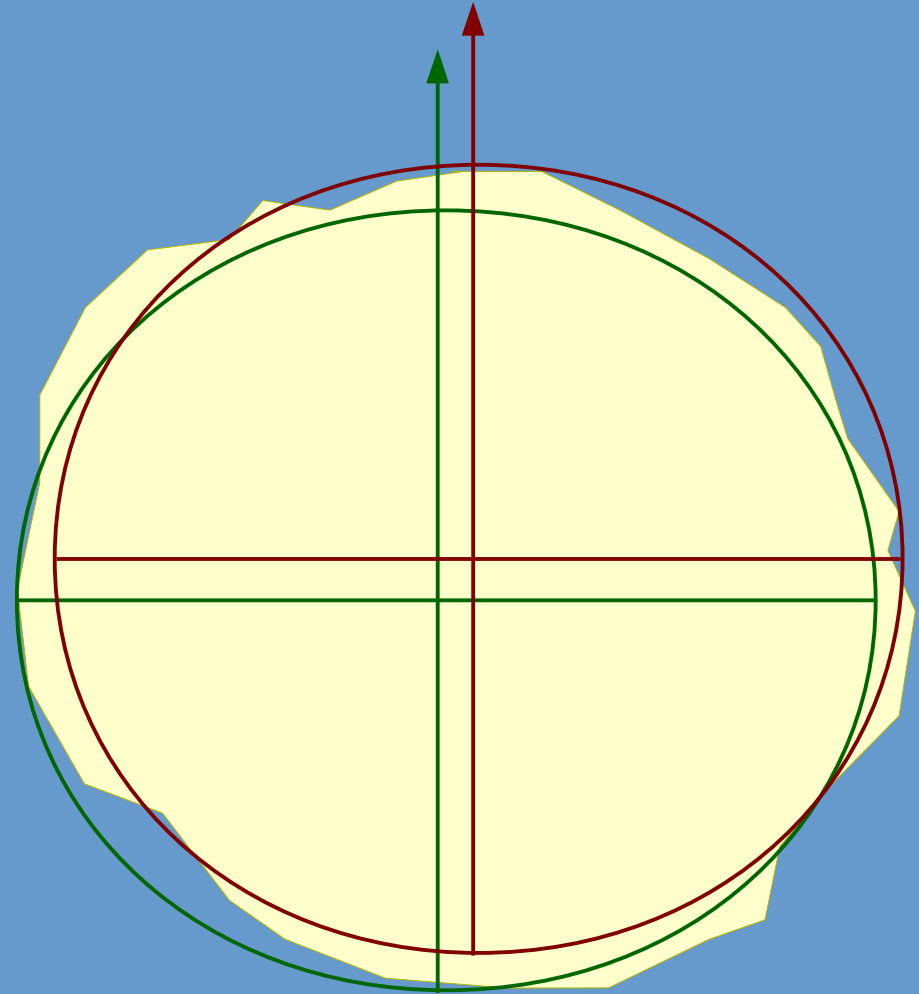


Datum

- Datum
 - Model of the size/shape of the Earth
 - Consists of an ellipsoid, defined by its geometry
 - Fit to the Earth (origin location)
 - Many are created for use only in specific areas of the earth
 - i.e., ellipsoid is fitted to only a portion of the earth and the center of the Earth is shifted in each
 - Ellipsoid is referenced to a point on the Earth surface
 - World Geodetic System (WGS84) is referenced to the center of Earth and designed for use globally

Datums

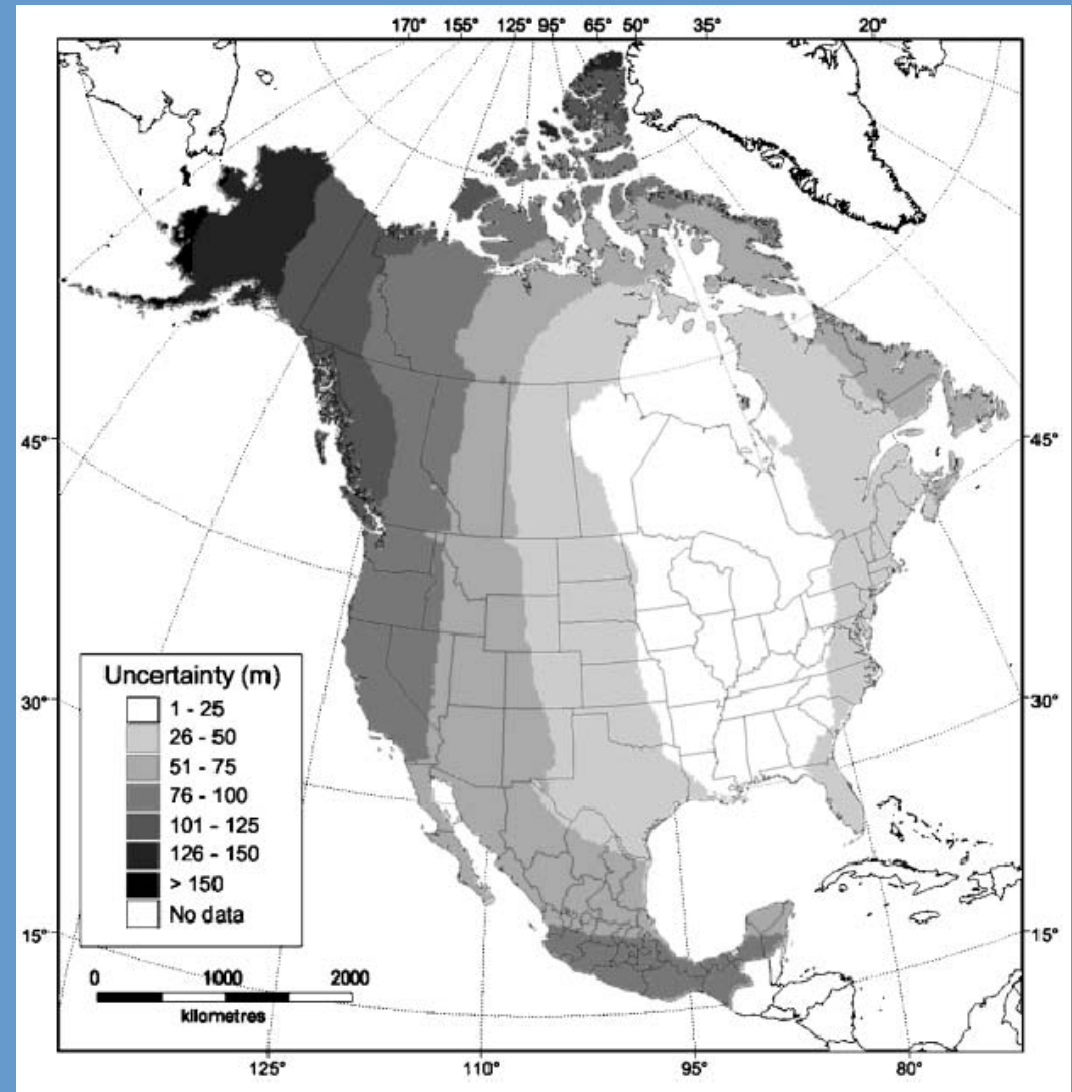
- NAD 27
 - Clarke 1866 ellipsoid
 - Best fit to North America
- NAD 83 – WGS 84
 - GRS80 ellipsoid
 - Origin is in the center of Earth's mass
 - Best fit to globe



Datums

- Consequences of using wrong datum?
 - Choice of datum affects location measurements in any projection
 - Can have serious consequences

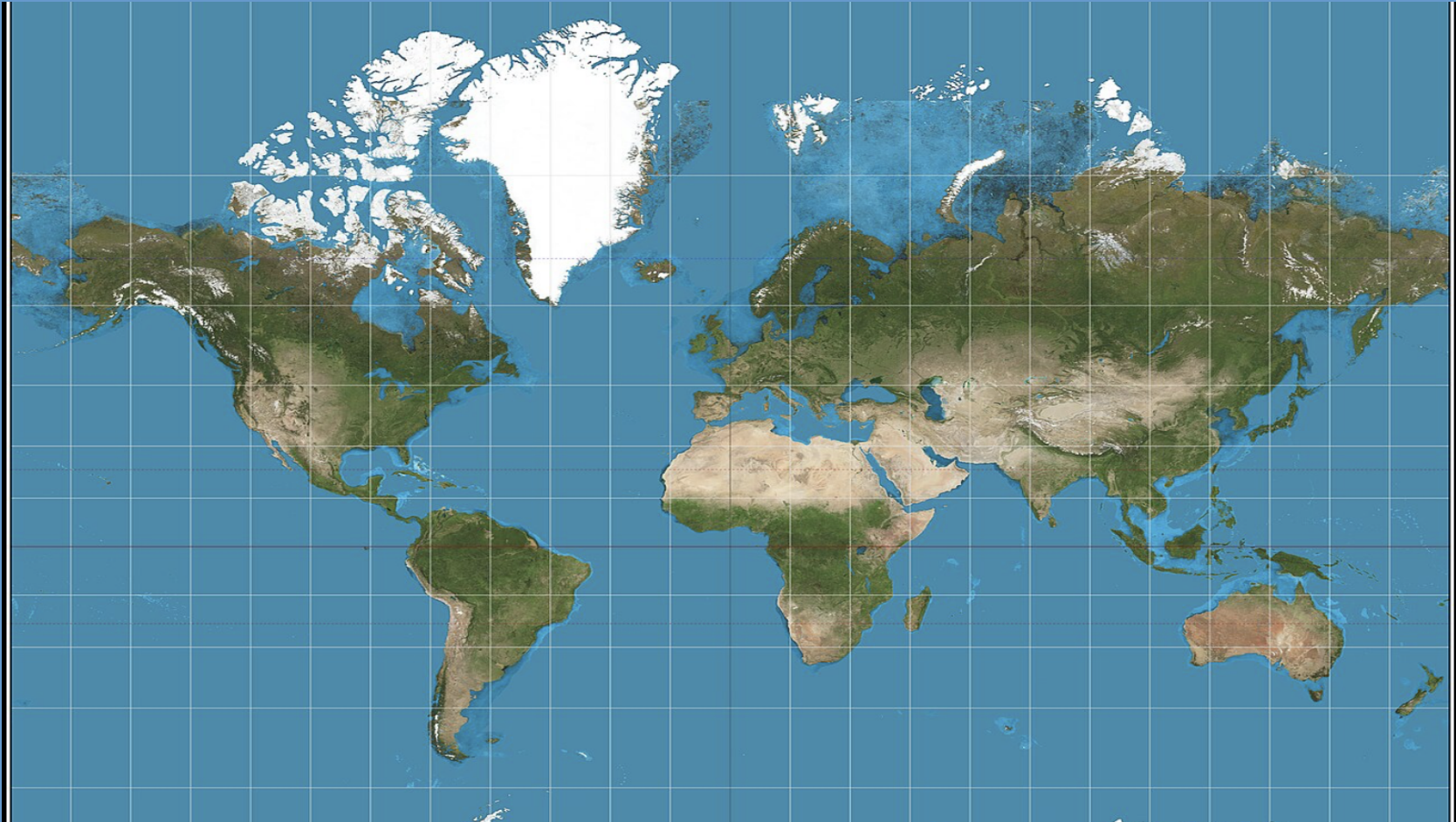
Distance shift for NAD 27 vs. NAD 83



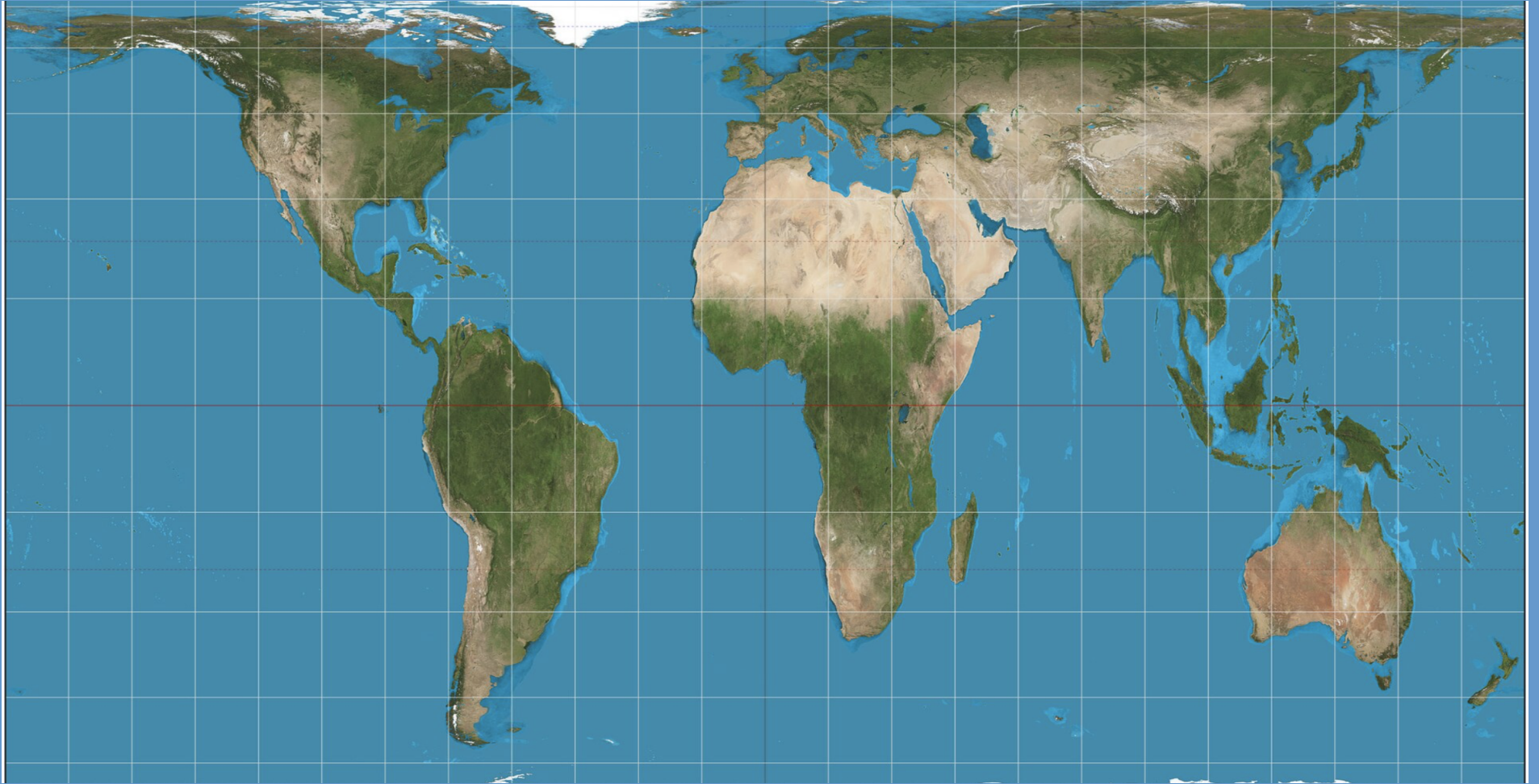
Projections

- ...or, Projected Coordinate System
- Mathematically defined conversions from location on 3-D Earth to 2-D space
 - Use geometric analogy of light source shining through clear Earth onto a “developable surface”
 - Distortion increases away from point of contact
- Geometric Families
 - Two most common developable surfaces
 - Cylindrical and Conic

Projections



Projections



Properties of Projections

- Properties of the Earth or features
 - Direction, shape, area, distance
- Projections distort some or all of these properties
 - Necessary because of the nature of moving from a 3D object to a 2D surface
 - Remember, projections are mathematical transformations from angular coordinates
 - No perfect transformation

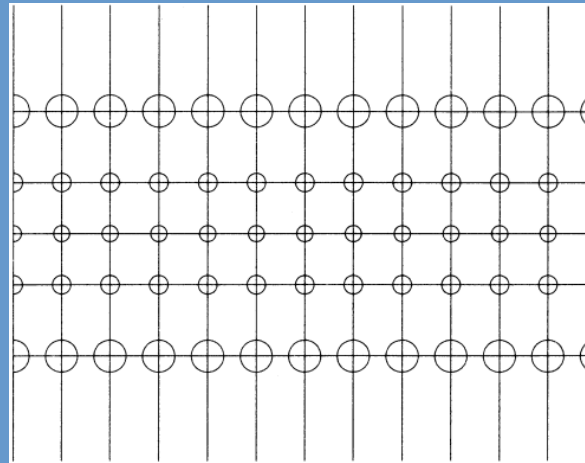
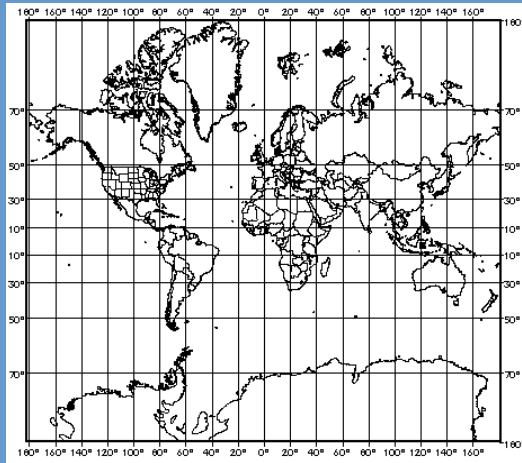
Projection Categories

- Categories are based on which property the projection preserves
 - Area, Equal Area
 - Shape (local), Conformal
 - Direction, Azimuthal
 - Distance, Equidistant
- Choosing a projection
 - Consider use of data or map
 - Consider geographic extent of analysis

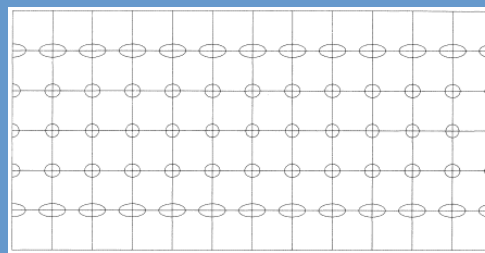
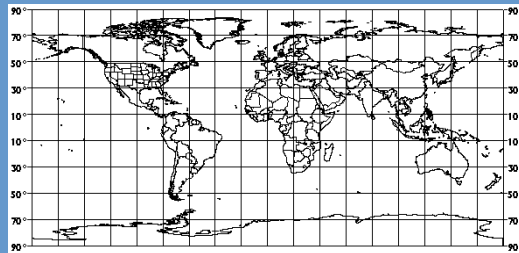
Projection Distortion

- Tissot Indicatrix
 - Graphically displays projection distortions

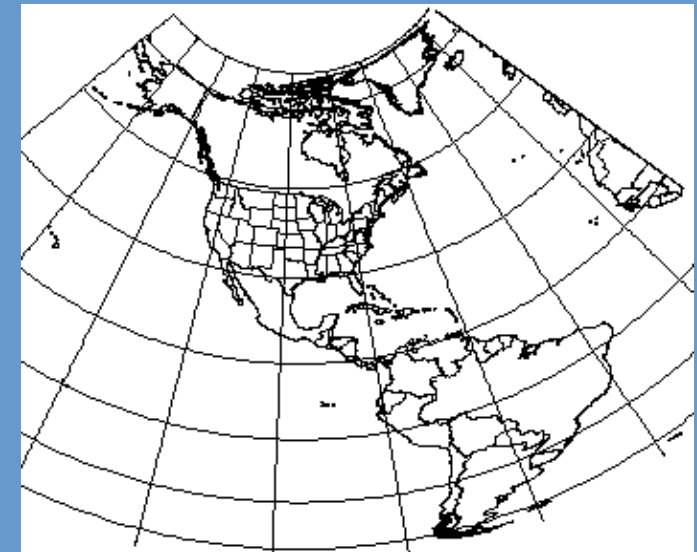
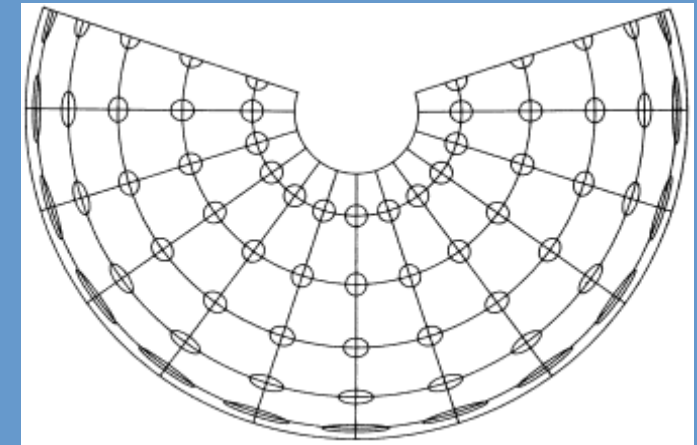
Mercator



Geographic (Lat/Long)



Albers Equal-Area



Geographic Coordinate Systems

- Geographic Coordinate Systems have:
 - A datum
 - An ellipsoid fit to the shape of the earth
 - An origin
 - Prime Meridian and Equator
 - Units
 - Angles: decimal degrees, degrees minutes seconds

Projected Coordinate Systems

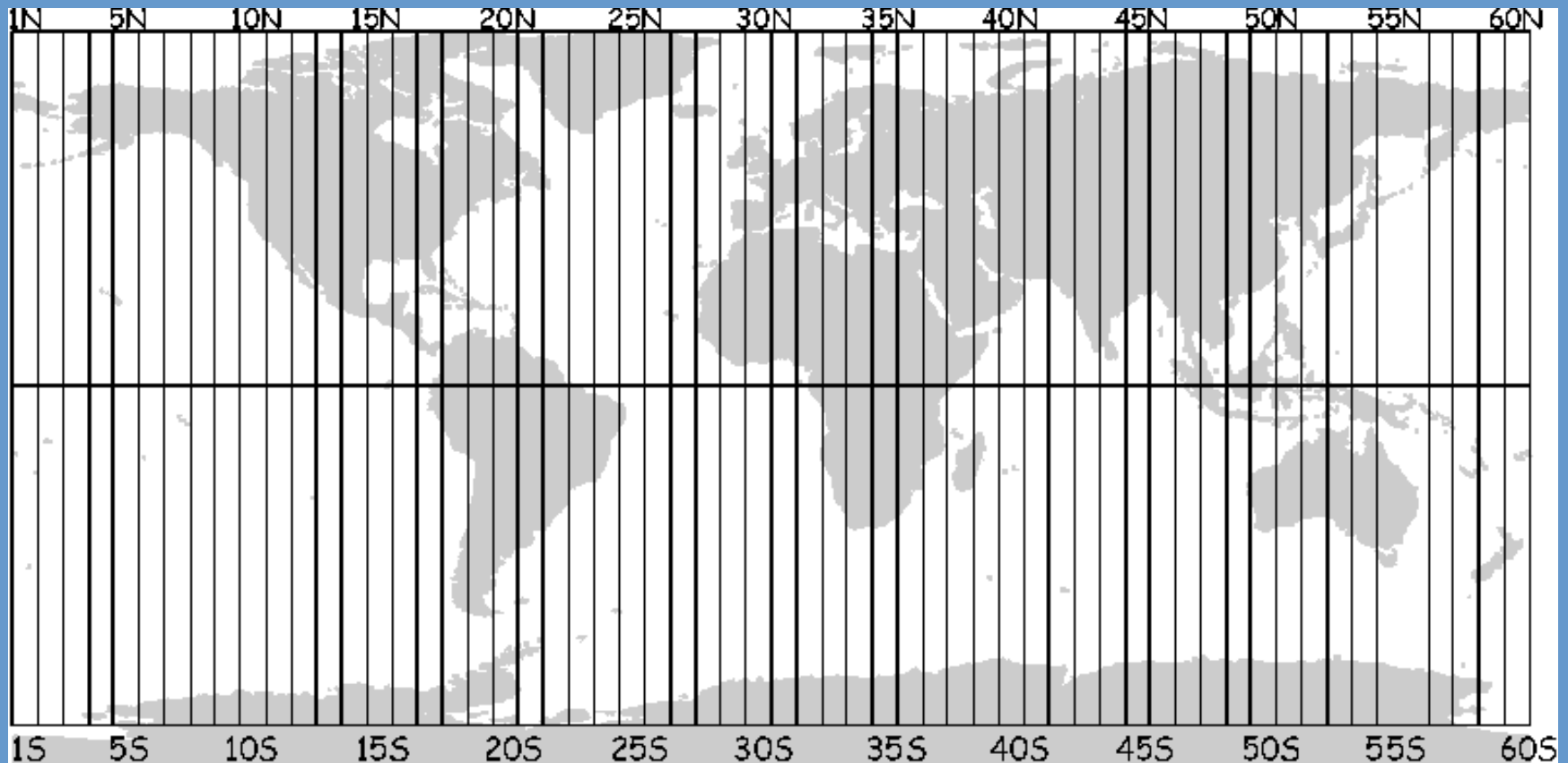
- Projected Coordinate Systems have:
 - A geographic coordinate system
 - Datum, origin, units
 - A projection
 - Developable surface
 - e.g., cylindrical, conic, other
 - An origin
 - $(x,y) = (0,0)$
 - Units
 - Distance: meters, feet

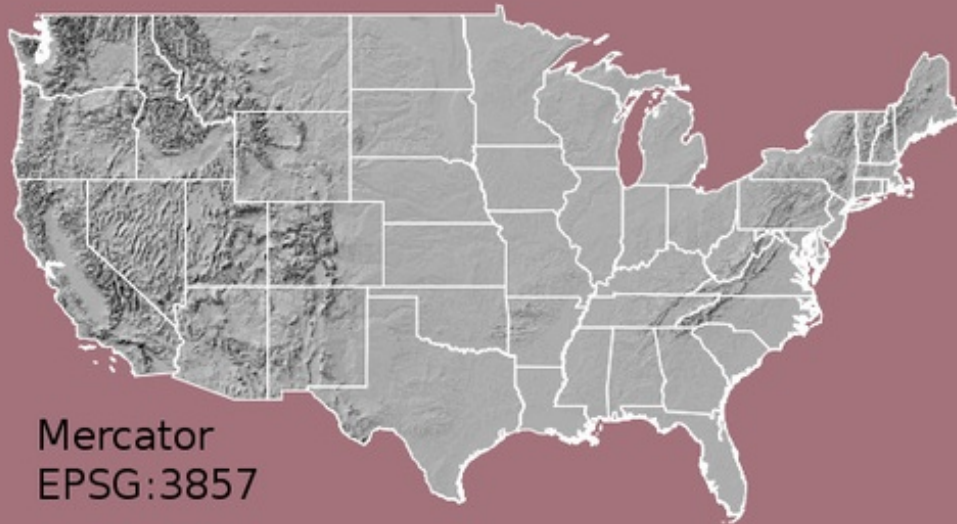
Commonly Used CS

- Geographic - aka Lat/Long
 - Uses angular units (e.g., degrees)
 - Can be used on GCS
 - Not projected - if mapped directly to 2-D representation is called "Plate Carree"
- Universal Transverse Mercator (UTM)
 - Transverse Mercator projection for each of 60 different zones, each 6 degrees of longitude wide
 - Essentially 60 different projections

Commonly Used CS

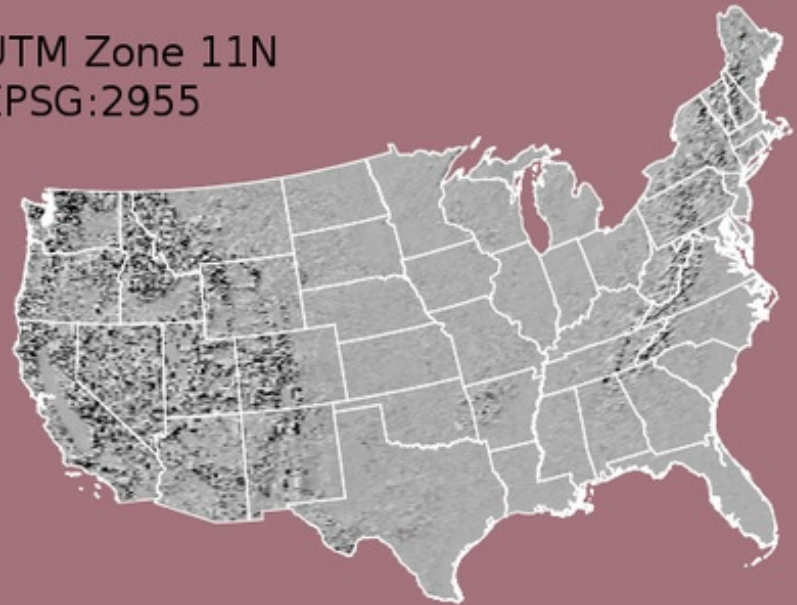
- Universal Transverse Mercator (UTM)





Mercator
EPSG:3857

UTM Zone 11N
EPSG:2955



U.S. National Atlas
Equal Area
EPSG:2163

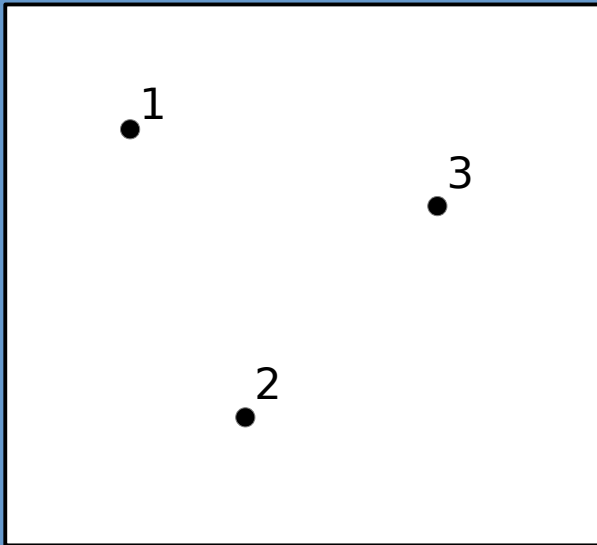


WGS 84
EPSG:4326

Managing Projections

- Feature locations (coordinates) stored in the “spatial” portion of spatial data
 - Coordinate system (projection) is defined and stored in .prj or other file

What we see on the screen is really just tabular information!



ID	X	Y
1	2	8
2	4	2
3	8	7

The correspondence between X,Y coordinates and locations on the Earth's surface is stored in the layer's .prj file

Managing Projections

- Important to understand the difference between the coordinate system of the data... and the coordinate system of the display
 - The locations of features in a spatial data layer are stored in a specific coordinate system
 - However, we can display the data in different coordinate systems

Managing Projections

- QGIS and projections
 - Coordinate system of the map panel (in QGIS project) is defined by first data source added
 - Can be changed at any time
 - Data added afterward will be reprojected “on the fly” to match coordinate system of data frame window
 - Use with caution, especially with large/complex datasets, because it slows down performance
 - DISPLAY ONLY, data are not changed

Managing Projections

- Good practice: normalize your data to same coordinate system
- Operations in GIS
 - Project (reproject)
 - Converts data from one coordinate system to another
 - Use when data currently have a defined coordinate system (e.g., geographic coordinates) and need to be converted to a different coordinate system
 - Creates a new spatial data layer, with updated location information (features) and .prj file

Managing Projections

- Good practice: normalize your data to same coordinate system
- Operations in GIS
 - Define Projection
 - Assigns projection information if there is none, or overwrites current projection information
 - Use when data have no coordinate system information or if it is wrong (but, only if you know the correct system)
 - Does not create new layer, spatial features are unchanged, creates/updates the .prj file